

MATH 3190-01 Fundamentals of Data Science Spring 2026

This syllabus contains information about this class. Please read this carefully and don't hesitate to ask me if you have questions.

Instructor:

Rick Brown

Office: Bristlecone Hall 214F

Email: richardbrown1@suu.edu

Class Times:

MWF 9:00 AM – 9:50 AM in Room 311 of Bristlecone Hall.

Office Hours:

Mondays: 10:00am – 12:00pm

Tuesdays: 10:30am – 11:15am

Wednesdays: 10:00am – 1:00pm

Thursdays: 10:30am – 11:15am and 2:00pm – 3:00pm

Fridays: 11:00am – 12:00pm

and by appointment. Email me any time to set up a time to meet!

You can also visit arrangr.com/brown/meetings to schedule a time.

Additionally, I can meet via Zoom during office hours if requested, so you may attend virtually.

Prerequisite:

C or better in 3150 or 3700 and a linear algebra class. In addition, coding experience is expected.

Canvas Page:

<https://suu.instructure.com/courses/1153375>

This site will contain all information in this syllabus plus more. Homework assignments and other information pertinent to this course will be posted on this web site, which will be updated frequently, so you should visit it regularly.

Optional Texts:

Each of these books is available online for free:

An Introduction to Statistical Learning with Applications in R, 2nd edition, By Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, New York: Springer, 2013.

Modern Data Science with R, 2nd edition, By Benjamin S. Baumer, Daniel T. Kaplan, Nicholas J. Horton, Chapman and Hall/CRC, 2021.

Introduction to Data Science: Data Analysis and Prediction Algorithms with R, 1st edition, By Rafael A. Irizarry, Chapman and Hall/CRC, 2020.

<https://rafalab.dfci.harvard.edu/dsbook/> {<https://rafalab.dfci.harvard.edu/dsbook/>}.
<https://rafalab.dfci.harvard.edu/dsbook/>.

Communication:

The primary way I will communicate with you outside of the classroom is through email. Please check your student email accounts daily.

Course Description:

This class will be an introduction to the mathematics and algorithms underlying the analytic techniques of data science. Topics covered will include regression and cross validation, gradient descent, dimension reduction, clustering and classification. Database management will also be discussed. Students will use a programming language and GitHub throughout this course.

Software:

We will be heavily utilizing **R** and RStudio in this class. **R** is free, open-source statistical software that is popular and used worldwide while RStudio is the most common IDE for **R**. If you prefer, you can use [Positron](#) as an alternative to RStudio. You can install **R** [here](#) and RStudio [here](#). We will also be using Git and [GitHub](#).

Class attendance and face-to-face designation:

- This class has a face-to-face designation and you are expected to attend class every day unless you are feeling ill. I will be recording class sessions and will post the videos on Canvas.
- If I feel ill, I will not come to campus and will instead teach the class from home or leave a video to watch. I will try my best to inform you the day before class if I will not be teaching in person. The link to view these classes will be on Canvas.

Assessment:

20% Homework and Labs
50% Weekly Quizzes
30% Final Project

Grade Scale

A	A-	B+	B	B-	C+	C	C-	D+	D	D-	F
≥ 93%	90%	87%	83%	80%	77%	73%	70%	67%	63%	60%	< 60%

Homework:

Please take the homework seriously and think of each one as writing a report. There will be typically be, on average, a little less than one homework assignment per week. You may work with classmates on homework assignments, but you must create your own solutions in your own words. Most homework should be completed using Quarto. You'll submit your homework on GitHub and the outputted html file on Canvas (to unlock the solutions). The lowest homework score will be dropped.

In addition to the written homework submission, you will also be required to submit a self-critique comment. The idea is that you should submit the

assignment, thoroughly look over the solutions, and then indicate in the comment anything you noticed that was incorrect or anything you have questions about. The homework is not considered submitted until you leave your comment.

Late homework will generally be accepted up to one day late, but at a **50%** grade reduction.

Labs:

Labs are assignments done (or at least started) in class. You will be expected to finish the lab before the next class period in most cases. There will not be quizzes on lab assignments.

Quizzes:

After nearly all homework assignments, there will be a quiz for you to take in the testing center. The quiz will feature a few questions similar to those you had to answer on the homework for that week. If you understand the homework, these quizzes should go smoothly. These quizzes will not focus on coding and will instead focus on interpretation, reading **R** code output, and occasionally answering more theoretical questions that you will explore in the homework. The lowest quiz score will be dropped.

Final Project:

There will be a group project toward the end of the semester where each group must research a data science/machine learning algorithm, learn the math and theory of it, apply it to a data set, write a report, and give a presentation to the class. Presentations will take place during the final period and the two or three classes period before, depending on the number of students enrolled in the class. Presentations should be about 40 minutes in length (this may change depending on the number of students in the class). Collaboration should take place through GitHub. More information on the project will come later in the semester.

SOME IMPORTANT DATES (see [here](#) for more):

Jan. 7	(Wednesday):	First day Spring Semester instruction
Jan. 19	(Monday):	MLK Day (no classes)
Jan. 28	(Wednesday):	Last day to drop without a W
Feb. 16	(Monday):	Presidents' Day (no classes)
Mar. 9–13	(Mon-Fri):	Spring Break (no classes)
Mar. 11	(Wednesday):	Last day to withdraw with a W
Mar. 31	(Tuesday):	Festival of Excellence (no classes)
Apr. 17	(Friday):	Last class day

Workload:

In general, you should expect to work two hours outside of class for every hour in class. That would be 5-6 hours outside of class every week for this course, on average, since we have $3 * 50/60 = 2.5$ hours in class each week

Honesty:

Scholastic dishonesty will not be tolerated and will be prosecuted to the fullest extent. You are expected to have read and understood the current issue of the [student handbook](#) regarding student responsibilities and rights, and for the intellectual property policy, information about procedures, and what constitutes acceptable behavior. More information can be found online at <https://www.suu.edu/academic-misconduct>.

AI Statement:

The goal of this class is for your to learn this material. If using artificial intelligence can help you learn, then you may use it. Do not use it as a crutch or use it to completely do your assignments. Always attempt each problem yourself first. Students who over-rely on AI tend to do significantly worse on quizzes and projects.

Accommodation:

Students with medical, psychological, learning, or other disabilities desiring academic adjustments, accommodations, or auxiliary aids will need to contact the Southern Utah University Coordinator of Disability Resource Center (DRC), in Room 206F of the Sharwan Smith Center or phone (435) 865-8022. The DRC determines eligibility for and authorizes the provision of services. I will work with you and the DRC to provide an appropriate accommodation.

Course Fee:

There is a fee of \$3.75 to take this course (one of the lowest fees on campus).

Other Help:

I *strongly* encourage you to attend office hours or to contact me if you have difficulty with the material. My job is to help you understand this material as well as possible, and I am flexible with meeting times. In addition, there is free tutoring available at the tutoring center. More information can be found
Tutoring is available [here](#).

Other information:

More university statements can be found [here](#).